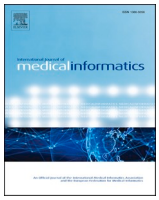
International Journal of Medical Informatics 202 (2025) 105975



Contents lists available at ScienceDirect

International Journal of Medical Informatics

journal homepage: www.elsevier.com/locate/ijmedinf



Review article



The accuracy of Machine learning in the prediction and diagnosis of diabetic kidney Disease: A systematic review and Meta-Analysis

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ARTICLE INFO

Keywords:
Diabetic kidney disease
Machine learning
Prediction
Diagnosis
Meta-analysis

ABSTRACT

Purpose: Machine learning (ML) has gained attention in diabetes management, particularly for predicting and diagnosing diabetic kidney disease (DKD). However, systematic evidence on its performance remains limited. This study evaluates the predictive and diagnostic accuracy of ML in DKD to support the development of tailored prevention strategies and non-invasive diagnostic tools.

Methods: A systematic search of PubMed, Embase, Web of Science, and Cochrane (up to April 14, 2024) identified relevant studies. Risk of bias was assessed using tools for predictive models, and *meta*-analysis included subgroup analyses based on task type, dataset, and model type.

Results: A total of 34 studies were included, with 19 on DKD risk prediction and 15 on diagnosis. For prediction, the pooled c-index was 0.81 (95% CI 0.79–0.83), sensitivity 0.81 (95% CI 0.74–0.86), and specificity 0.82 (95% CI 0.73–0.89). For diagnosis, the pooled c-index was 0.81 (95% CI 0.79–0.83), sensitivity 0.81 (95% CI 0.78–0.84), and specificity 0.75 (95% CI 0.72–0.79).

Conclusions: ML shows promising accuracy in DKD prediction and diagnosis, offering a viable tool for early screening and risk assessment.

1. Introduction

The Global Picture of Diabetes, published by the International Diabetes Federation in 2021, suggests that an estimated 537 million individuals aged 20–79 are living with diabetes. It is anticipated that this figure will increase to 783 million (12.2 % of the global population) by 2045[1]. Diabetes can damage target organs, such as the kidneys, resulting in the onset of diabetic kidney disease (DKD), which may ultimately progress to end-stage kidney disease (ESKD)[2]. The global proportion of ESKD patients attributable to diabetes has steadily increased, from 22.1 % in 2000 to 31.3 % in 2015[3]. This trend underscores that DKD has become a leading cause of ESKD, posing a significant healthcare burden[4]. Therefore, early prediction and diagnosis of DKD hold profound clinical significance.

The onset and progression of DKD are influenced by various pathophysiological mechanisms, and as a result, there is currently no definitive tool for predicting the risk of DKD. Consequently, it is challenging to formulate prevention strategies for high-risk populations. At present, the diagnosis of DKD primarily relies on renal biopsy[5]. However, because renal biopsy is an invasive procedure linked to possible complications like bleeding and infection[6], patient compliance is often suboptimal. Therefore, there remains a need to explore effective tools for the early prediction and diagnosis of DKD.

Machine learning (ML) has recently drawn increasing attention from clinicians [7], owing to its promising accuracy for disease progression, prognosis prediction, and diagnosis [8,9]. Researchers have begun to focus on using ML to develop risk prediction and diagnostic tools for DKD, and there is already a *meta*-analysis of machine learning applied to DKD, and this article is a single-region study that did not strictly differentiate between ML models used for prediction and diagnosis of DKD. However, in clinical practice, prediction and diagnosis of diseases have different meanings, so we strictly differentiated between ML for prediction and diagnosis and included relevant global studies to perform the *meta*-analysis.Therefore, this systematic review was undertaken to

Abbreviations: XGBoost, eXtreme Gradient Boosting; RF, random forest; DT, decision tree; LR, ogistic regression; KNN, K-nearest neighbor; SVM, support vector machine; EN, elastic net; GBDT, GradientBoosting Decision Tree; LightGBM, Light Gradient Boosting Machine; LDA, linear discriminant analysis; QDA, quadratic discriminant analysis; AdaBoost, Adaptive Boosting; NB, naïve bayes; KNN, K- neighbors; ANN, Artificial Neural Network; RBF network, Radial Basis Function Network; FNN, Fuzzy Neural Network; PRN, Pattern Recognition Network; CatBoost, Categorical Boosting.

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https://doi.org/10.1016/j.ijmedinf.2025.105975
Received 19 March 2025; Received in revised form 14 May 2025; Accepted 16 May 2025
Available online 17 May 2025
1386-5056/© 2025 Published by Elsevier B.V.

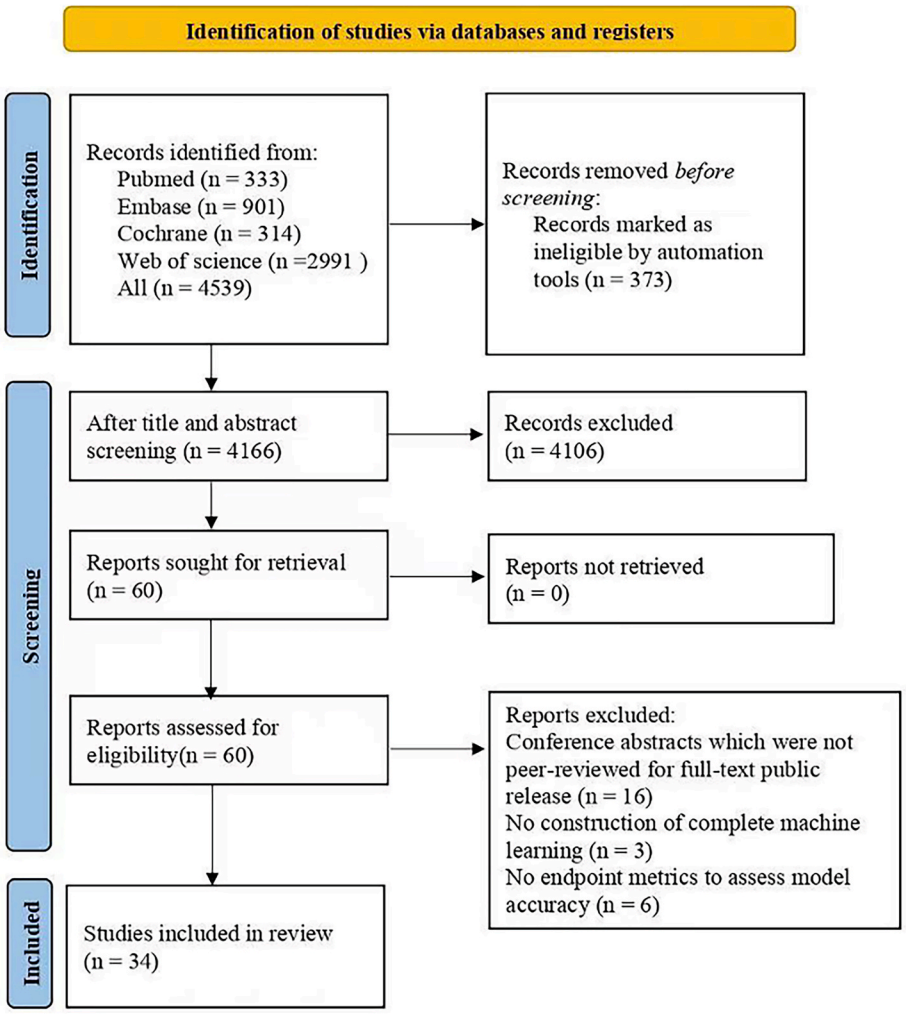


Fig. 1. The process of literature selection.

assess the efficacy of ML in the prevention and diagnosis of DKD, providing evidence-based insights for the future development of predictive and diagnostic tools.

2. Methods

2.1. Study Registration

This study was performed according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [10] and prospectively registered on the PROSPERO website (Registration No.: CRD42024565987).

2.2. Eligibility criteria

2.2.1. Inclusion Criteria

(1) The eligible study types were case-control, cohort, and cross-sectional studies.

(2) Studies comprehensively developed ML models for DKD prediction or diagnosis.

(3) Although some studies may lack an independent validation set, we acknowledge the contribution of these studies and have included them in this systematic review.

(4) They were published in English.

2.2.2. Exclusion Criteria

(1) Full-text conference abstracts were not peer-reviewed.

(2) The studies performed only risk factor analysis without constructing a complete ML model.

(3) Studies did not provide any of the following outcome metrics for evaluating the accuracy of the ML model:: c-index, sensitivity, specificity, accuracy, precision, confusion matrix, F1 score, as well as calibration curve.

(4) They focused solely on univariate analysis for prediction accuracy.

2.3. Data sources and search strategy

PubMed, Embase, Cochrane Library, and Web of Science databases were systematically searched with the literature time frame restricted from the respective database inception dates to April 14, 2024. MeSH terms and free text keywords were collectively used for search, without any geographic limitations. We used “Diabetic Nephropathies” and “machine learning” as subject terms in PubMed, Cochrane, and Web of Science databases, while employing “diabetic nephropathy” and “machine learning” as subject terms in the Embase database. Detailed search strategies are presented in Table S1..

2.4. Study selection and data extraction

The retrieved documents were uploaded to Endnote, and titles and

Table1
The basic characteristics of the included studies.

No.	First Author	Year of Publication	Country of the Author	Type of Research	Source of Patients	Task Type	Number of cases of DKD	Total number of cases	Total number of cases in the training set.	Method of generating the validation set	Number of cases in the validation set.	Methods for Handling Missing Data	Types of Models Used
1	Yin, J. M	2024	China	Cohort Study	single-center	Forecast	119	562	562	10-fold cross	/	Delete	XGBoost, RF DT, LR
2	Xing, Y. W	2024	China	Case-control	single-center	Diagnosis	199	359	359	5-fold cross	/	The Random Forest method (RF)	XGBoost
3	Wang, X	2024	China	Cohort Study	single-center	Forecast	59	403	/	bootstrap	/	Not mentioned	LR
4	Ma, J. J	2024	China	Cohort Study	database	Forecast	Not mentioned	8610	2412	Random Sampling External Validation	set1:1981 set2:3183	Deletion and Multiple Imputation Method	RF
5	Jiang, C. X	2024	China	Cohort Study	single-center	Forecast	248	670	/	/	/	Not mentioned	LR
6	Cai, S. S	2024	China	Case-control	single-center	Diagnosis	74	210	147	Random Sampling	63	Not mentioned	LR, DT, RF
7	Betzler, B. K	2024	Singapore	Cohort Study	database	Diagnosis	/	/	13,284	External Validation	set1:1969 set2:712	Delete	LR
8	Zhao, Y. T	2023	China	Case-control	multicenter	Diagnosis	295	678	202	Random Sampling External Validation	set1:87 set2:373	Delete	LR
9	Tan, H. Z	2023	Singapore	Case-control	single-center	Diagnosis	65	102	/	bootstrap	/	Delete	LR
10	Su, X. E	2023	China	Case-control	single-center	Diagnosis	105	219	175	Random Sampling	44	Not mentioned	KNN, SVM, LR
11	Shi, S. M	2023	China	Case-control	single-center	Diagnosis	149	515	370	Random Sampling	155	Multiple Imputation Method	RF
12	Sarkhosh, S. M. H	2023	Iran	Cohort Study	database	Forecast	1396	3450	1907	Prospective validation	1543	Mean value	RF, SVM, DT, LR, XGBoost
13	Sabanayagam, C	2023	Singapore	Cohort Study	multicenter	Forecast	162	1365	/	Random Sampling	/	Imputed Mean Value	EN, GBDT, LASSO, XGBoost, RF
14	Mu, X. D	2023	China	Cohort Study	single-center	Forecast	127	521	365	Random Sampling	156	Not mentioned	Nomogram model
15	Liu, X. Z	2023	China	Case-control	multicenter	Diagnosis	1856	3624	2899	Random Sampling	725	Multiple Imputation Method	CatBoost, LightGBM, XGBoost, GBDT, RF, LDA, LR, AdaBoost, NB, KNN, DT, SVM
16	Liao, L. N	2023	China	Cohort Study	single-center + database	Forecast	388	1526	1019	Random Sampling External Validation	set1:507 set2:3899	Not mentioned	LR
17	Cao, X	2023	China	Cohort Study	single-center	Forecast	188	597	418	Random Sampling	179	Delete	LR
18	Dong-mei Zhou	2022	China	Case-control	multicenter	Diagnosis	52	102	102	10-fold cross	/	Not mentioned	LR

(*continued on next page*)

Table1 (*continued*)

No.	First Author	Year of Publication	Country of the Author	Type of Research	Source of Patients	Task Type	Number of cases of DKD	Total number of cases	Total number of cases in the training set.	Method of generating the validation set	Number of cases in the validation set.	Methods for Handling Missing Data	Types of Models Used
19	Zhang, W. G	2022	China	Case-control	single-center	Diagnosis	477	1258	929	Random Sampling	329	Mean value	RF, SVM
20	Yang, J	2022	China	Cross-sectional study	single-center	Forecast	195	706	521	Prospective validation	185	Delete	LR
21	Sun, L	2022	China	Cohort Study	single-center	Forecast	2087	14,628	/	Random Sampling	4876	Not mentioned	LR
22	Hosseini Sarkhosh, S. M	2022	Iran	Cohort Study	single-center	Forecast	2210	6235	/	Random Sampling	/	Deletion + Mean + Median	DT, LR,SVM, XGBoost, RF
23	Dong, Z. Y	2022	China	Cohort Study	single-center	Forecast	408	816	652	Random Sampling	164	Deletion + RF Method	LightGBM, XGBoost, AdaBoost, ANN, DT, SVM, LR
24	Allen, A	2022	the United States	Cohort Study	multicenter	Forecast	/	111,046	62,994	Random Sampling External Validation	set1:7656 set2:23073	Mean value	XGBoost, RF
25	Afrash, M. R	2022	Iran	Cross-sectional study	single-center	Forecast	150	327	/	10-fold cross	/	Not mentioned	RBF network, FNN, PRN, SVM, DT, KNN
26	Xi, C. F	2021	China	Cohort Study	single-center	Forecast	203	1095		bootstrap	/	Not mentioned	LR
27	Wang, G. Y	2021	China	Cross-sectional study	single-center	Forecast	313	2163	/	10-fold cross	/	Delete	LR
28	Maniruzzaman, M	2021	Bangladesh	Case-control	database	Diagnosis	115	266	133	External Validation	133	Not mentioned	LDA, SVM, LR, KNN, NB, ANN
29	Jiang, K	2021	China	Case-control	single-center	Diagnosis	193	535	/	/	/	Mean value + deletion	DT, RF
30	Shi, R	2020	China	Cross-sectional study	multicenter	Forecast	1597	4219	/	/	/	Delete	LR
31	Kitamura, S	2020	Japan	Case-control	single-center	Diagnosis	31	83	/	Random Sampling	/	Not mentioned	DL
32	Jiang, S. M	2020	China	Case-control	single-center	Diagnosis	143	302	214	Random Sampling	/	Not mentioned	LR
33	Hu, Y. H	2020	China	Cohort Study	multicenter	Forecast	701	3489	/	/	/	Delete	LR
34	Yang, Z. H	2019	China	Case-control	single-center	Diagnosis	83	213	/	/	19	Not mentioned	LR

Abbreviations: XGBoost, eXtreme Gradient Boosting; RF, random forest; DT, decision tree; LR, Logistic regression; KNN, K–nearest neighbor; SVM, support vector machine; EN, elastic net; GBDT, GradientBoosting Decision Tree; LightGBM, Light Gradient Boosting Machine; LDA, linear discriminant analysis; QDA, quadratic discriminant analysis; AdaBoost, Adaptive Boosting; NB, naïve bayes; KNN, K Nearest Neighbors; ANN, Artificial Neural Network; RBF network, Radial Basis Function Network; FNN, Fuzzy Neural Network; PRN, Pattern Recognition Network; CatBoost, Categorical Boosting.

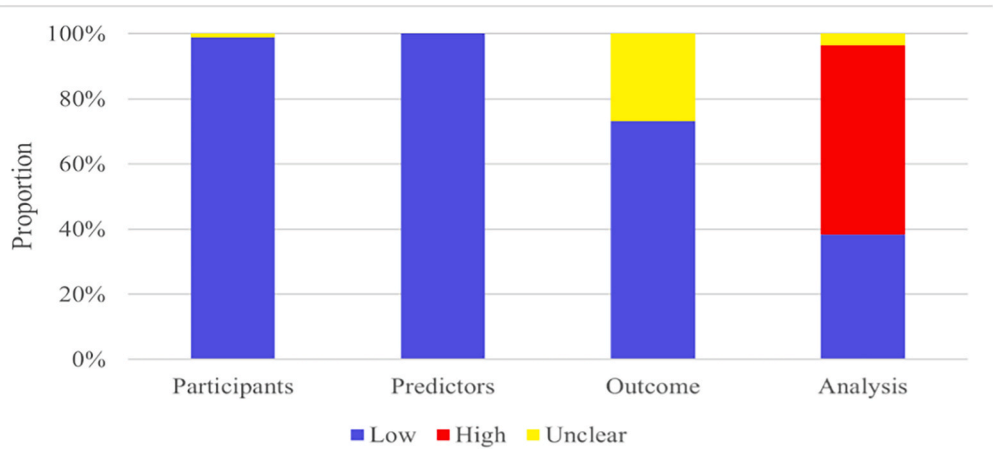


Fig. 2. Summary of PROBAST risk of bias.

abstracts were reviewed after the removal of duplicates. The full texts of the remaining studies were downloaded and read to identify eligible studies. Before carrying out data extraction, a standard spreadsheet was created for data extraction. Extracted information included title, authors, year of publication, country of authors, study type, source of patients, sort of task, diagnostic criteria for DKD, length of follow-up, number of DKD cases, total number of cases and number of DKD cases in the training set and the validation set, method of generating the validation set, methods of avoiding overfitting, and approaches of dealing with missing values, variable/feature screening method, type of model used and modeling variables. The literature screening and data extraction were performed independently by two researchers. Cross-checking was performed after completion. If a dispute arose, a third researcher (Maojun Chen) helped to solve it.

2.5. Risk of bias in studies

The risk of bias of the encompassed original studies was evaluated through Prediction model Risk Of Bias ASsessment Tool (PROBAST) [11], which contained a large number of questions in four different domains: participants, predictors, outcomes, and statistical analysis, reflecting the overall risk of bias and overall applicability. The domains had 2, 3, 6, and 9 questions respectively with three responses (yes/probably yes, no/probably no, and no information). A domain was deemed to be at high risk if at least one question was answered by no or probably no. A domain was considered low risk if all questions were answered with yes or possibly yes. The overall risk of bias was rated as low when all domains were assessed as low risk, and it was deemed high if at least one domain was classified as high risk.

Two researchers (Changmao Dai and Xiaolan Sun) independently conducted the risk of bias assessment based on PROBAST, and cross-checked their work. A third researcher (Fang Ding) was asked to assist with adjudication if there was a dispute.

2.6. Synthesis methods

A *meta*-analysis of the c-index, which was used to assess the overall accuracy of ML models, was conducted. In certain original studies, when the c-index was missing 95 % CI and standard error (SE), the study of Debray TP et al.[12] was referred to estimate their SE. The heterogeneity was assessed among studies using the I^2 index. When $I^2 > 50$ %, a random-effects model was employed; otherwise, a fixed-effects model was leveraged.

Moreover, bivariate mixed-effects models were utilized for the *meta*-analysis of sensitivity and specificity. Since most original studies did not report the diagnostic 2x2 contingency table, the table was reconstructed by combining case numbers with sensitivity, specificity, and precision. The *meta*-analysis was completed via STATA 15 (StataCorp, College Station, TX, USA).

3. Results

3.1. Study selection

Based on the retrieval strategy developed, a total of 4,539 articles were systematically searched, with no extra studies identified by reviewing the reference lists. After 373 duplicates were ostracized, 4,166 articles were screened based on titles and abstracts. Among these, 4,106 were removed for not satisfying the inclusion criteria. The full texts of the remaining 60 studies were checked. Finally, 34 studies were selected in our *meta*-analysis [13–45]. The literature selection process is illustrated in Fig. 1.

3.2. Study characteristics

This *meta*-analysis encompassed 34 studies published between 2019 and 2024, covering six countries. Of these, 25 studies were from China [13–18,20,22,23,26–33,35,38,39,41,43–45], three studies were from Singapore [19,21,25], one from the United States [36], three from Iran [24,34,37], one from Bangladesh [40], and one from Japan [42]. In total, the analysis involved 170,894 participants. 19 studies [13,15–17,24–26,28,32–37,46,47] focused on predicting the risk of DKD, of which 15 studies[13,15–17,24–26,28,33–36,38,44,46] were cohort studies, and four studies[32,37,39,41] were cross-sectional studies. There were 15 articles[14,18–23,27,30,31,40,42,43,45,48] on the diagnosis of DKD, of which 14 studies [14,18,20–23,27,30,31,40,42,43,45,48] are case-control studies, and one study[19] was a cohort study. Among the included articles, 22 were single-center studies, 7 were multi-center studies, and 5 used data from registries or datasets. In 8 articles, the diagnostic criterion for DKD was renal biopsy; 18 articles primarily employed glomerular filtration rate or creatinine clearance rate reduction and microalbuminuria as diagnostic criteria, and 8 articles did not mention diagnostic standards. Additionally, 20 articles reported methods for handling missing data, with deletion primarily used in 12 studies. The included studies encompassed 23 models(XGBoost, RF, DT, logistic regression [LR], ResNet, KNN, SVM, LASSO, GBDT, Catboost, LightGBM, Extra Trees, LDA, QDA, Adaboost, NB, Ridge Classifier, artificial neural network, pattern recognition network, radial basis function network, FNN, ANN, Deep Learning). The LR algorithm was predominantly used in studies on both prediction and diagnosis tasks. The basic information is presented in Table 1.

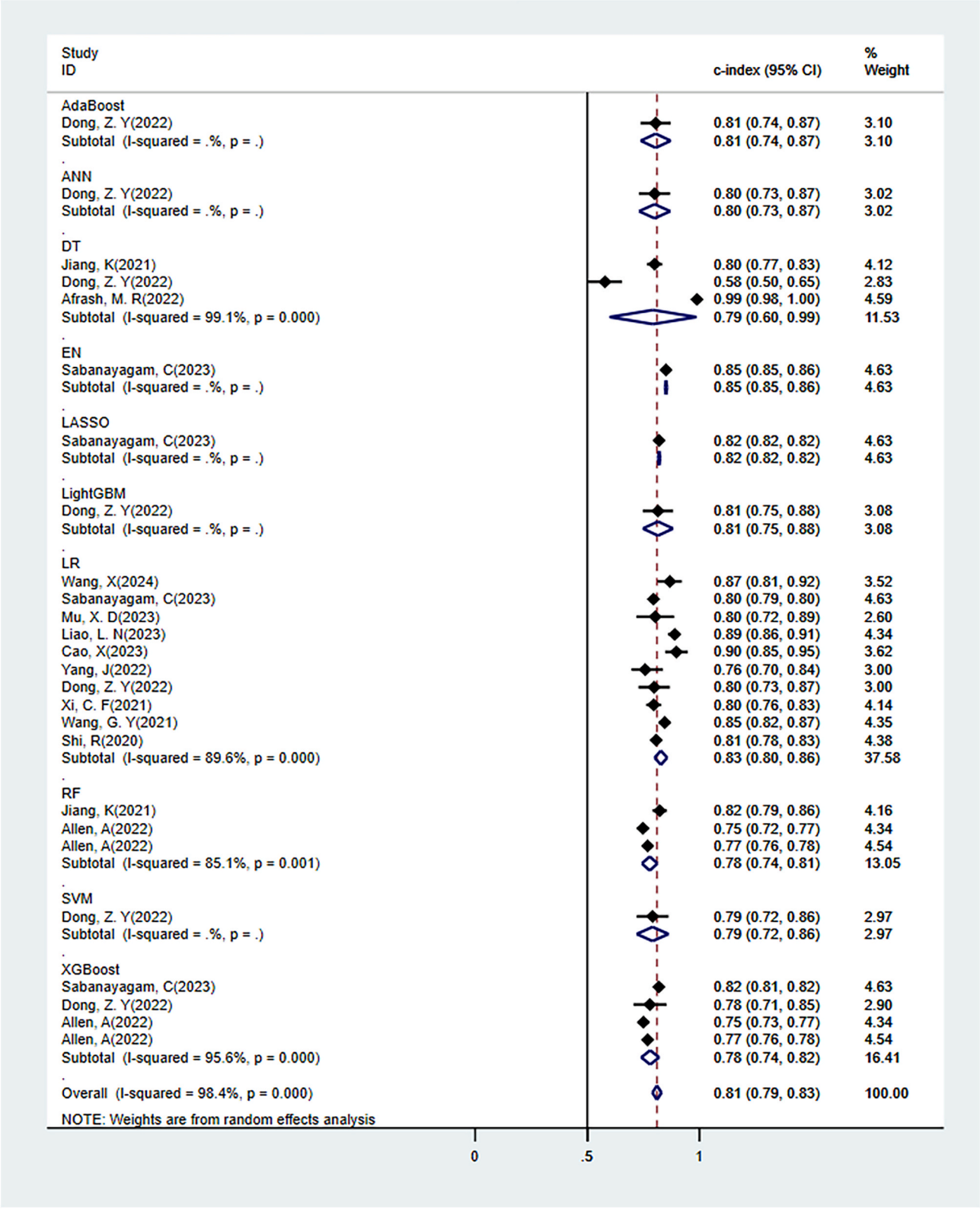


Fig. 3. The forest plot of the c-index for the models in predicting the risk of DKD in the validation set.

3.3. Risk of bias in studies

A risk of bias assessment was conducted for the 86 included models. One model was derived from a retrospective cohort study, and therefore, its data source was regarded as carrying a high risk of bias. 26 models had an Events Per Variable (EPV) < 10 or a validation sample size < 100, resulting in a high risk of bias in the statistical analysis. A total of 21 models directly excluded missing data or did not specify the method for handling missing data. These models failed to apply appropriate techniques for managing missing values in the included subjects and were therefore classified as high risk for bias. Similarly, 22 models were categorized as high risk due to the lack of consideration for overfitting

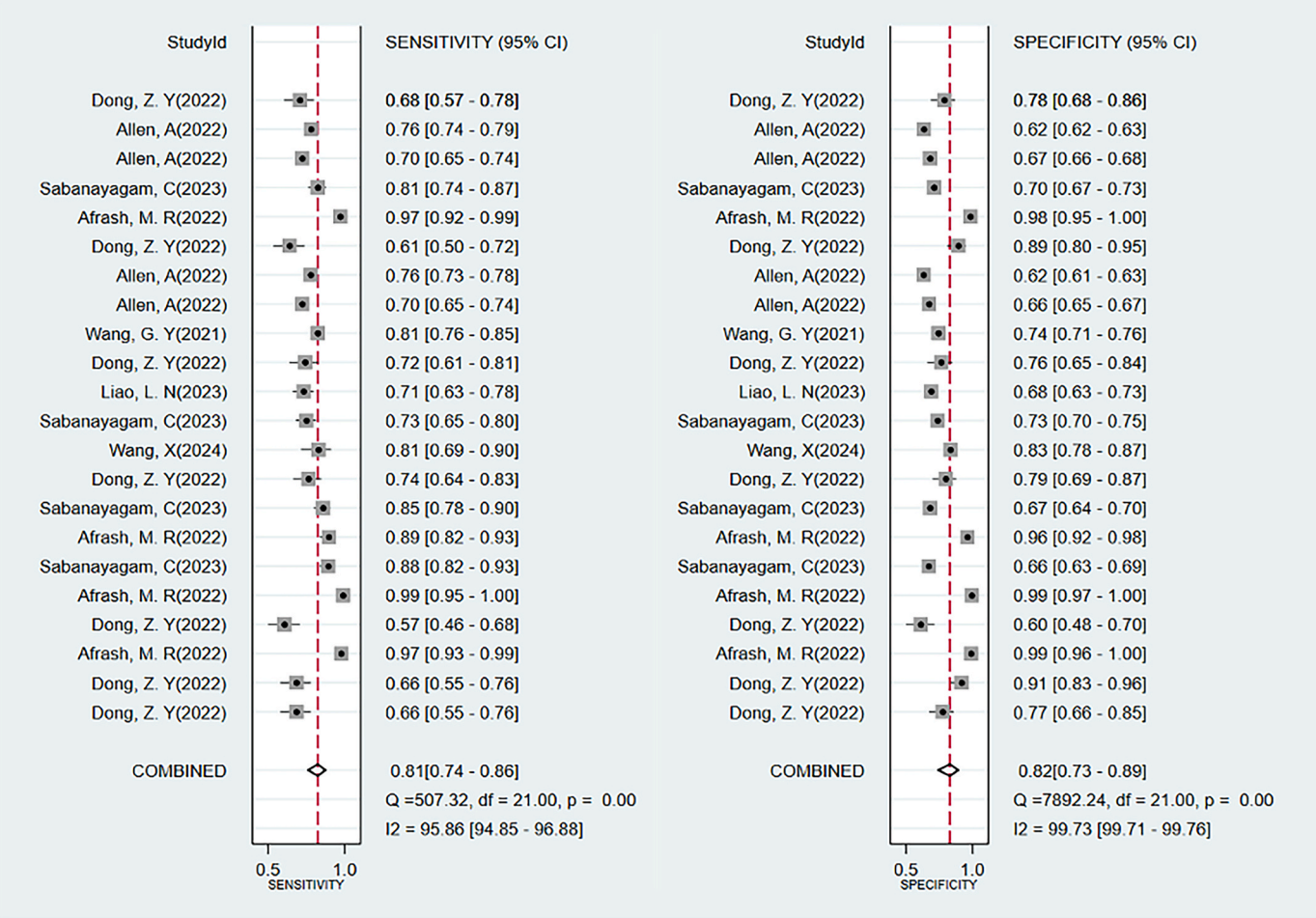


Fig. 4. The forest plot of sensitivity and specificity for the models in predicting the risk of DKD in the validation set.

Table 2
The summary c-index of the models in predicting the risk of DKD.

No.	Model	Training Set Number	Sample size	C-index(95 %CI)	Validation Set Number	Sample size	C-index(95 %CI)
1	ANN				1	164	0.800(0.730–0.869)
2	Adaboost				1	164	0.805(0.738–0.872)
3	DT	2	2469	0.769(0.686–0.852)	3	1571	0.793(0.600–0.986)
4	LightGBM				1	164	0.815(0.747–0.882)
5	XGBoost	2	2469	0.878(0.704–1.051)	4	32,258	0.780(0.740–0.820)
6	LR	10	23,555	0.826(0.773–0.878)	10	7155	0.828(0.800–0.856)
7	RF	4	11,116	0.872(0.822–0.922)	3	1080	0.778(0.744–0.813)
8	SVM	1	1907	0.747(0.712–0.782)	1	164	0.791(0.720–0.862)
9	EN				1	1365	0.851(0.847–0.856)
10	LASSO				1	1365	0.820(0.816–0.825)
11	Overall	20	41,516	0.832(0.803–0.862)	26	76,179	0.810(0.790–0.831)

and underfitting in the prediction models. In total, 50 models were identified as having a high risk of bias (Fig. 2), primarily related to statistical analysis.

3.4. Meta-analysis

3.4.1. Prediction task

3.4.1.1. Synthesized results. In the included studies on the prediction task, the c-index of 20 models in the training set was reported, involving a total of 41,516 cases. After summarizing the data using a random-effects model, the c-index was 0.832 (95 % CI: 0.803–0.862), with a sensitivity of 0.79 (95 % CI: 0.74–0.84) and specificity of 0.79 (95 % CI: 0.74–0.83). The studies on the prediction task also reported 26 models in the validation set, which included a total of 76,179 patients. Upon summarizing the data via a random-effects model, the c-index, sensitivity, and specificity were 0.810 (95 % CI: 0.790–0.831), 0.81 (95 % CI: 0.74–0.86), and 0.82 (95 % CI: 0.73–0.89), respectively. The results are detailed in Fig. 3, Fig. 4, Table 2, and Table 3.

3.4.1.2. Subgroup analysis. Given the differences in detection performance across various models, a subgroup analysis was carried out based on the type of models. The results demonstrated that LR was the predominant model type. In the training cohort, the pooled c-index for LR was 0.826 (95 % CI: 0.773–0.878), with a sensitivity of 0.78 (95 % CI: 0.72–0.82) and specificity of 0.77 (95 % CI: 0.71–0.82). In the validation

Table 3
The summary sensitivity and specificity of the models in predicting the risk of DKD.

No	Model	Training Set Number	sen(95 %CI)	spe(95 %CI)	Validation Set Number	sen(95 %CI)	spe(95 %CI)
1	ANN				2	0.66–0.98	0.91–0.99
2	Adaboost				1	0.66	0.77
3	DT	2	0.68–0.89	0.66–0.89	2	0.58–0.99	0.60–0.99
4	LightGBM				1	0.74	0.78
5	XGBoost	2	0.7–0.86	0.72–0.88	4	0.74(0.69–0.78)	0.68(0.63–0.72)
6	LR	8	0.78(0.72–0.82)	0.77(0.71–0.82)	5	0.77(0.71–0.81)	0.74(0.69–0.78)
7	RF	3	0.68–0.92	0.74–0.90	2	0.7–0.76	0.62–0.66
8	SVM	1	0.72	0.65	2	0.61–0.97	0.87–0.98
9	EN					0.88	0.65
10	LASSO					0.84	0.67
11	Overall	16	0.79(0.74–0.84)	0.79(0.74–0.83)	22	0.81(0.74–0.86)	0.82(0.73–0.89)

cohort, a random effects model was used, and the pooled c-index for LR was 0.828 (95 % CI: 0.800–0.856), with a sensitivity of 0.77 (95 % CI: 0.71–0.81) and specificity of 0.74 (95 % CI: 0.69–0.78). The analysis results demonstrated that LR was not inferior to other ML models (Tables 2-3).

3.4.2. Diagnosis task

3.4.2.1. Synthesized results. In the studies on the diagnosis of DKD, there were 11 models in the training set, with 3,575 cases. The c-index, sensitivity, and specificity were 0.82 (95 % CI: 0.73–0.89), 0.84 (95 % CI: 0.81–0.87), and 0.87 (95 % CI: 0.82–0.91), as summarized by a random-effects model. There were 40 models in the validation set, incorporating a total of 59,795 patients. The c-index, sensitivity, and specificity were 0.812 (95 % CI: 0.794–0.829), 0.81 (95 % CI: 0.78–0.84), and 0.75 (95 % CI: 0.72–0.79), respectively, as calculated by a random-effects model. Specific results can be found in Figs. 5-6 and Tables 4-5.

3.4.2.2. Subgroup analysis. Due to differences between various diagnostic models, subgroup analyses were conducted based on the model types. The results showed that LR was the most prevalent model type, followed by ResNet. In the training cohort, the pooled c-index for LR was 0.904 (95 % CI: 0.873–0.935), with a sensitivity of 0.84 (95 % CI: 0.75–0.90) and specificity of 0.83 (95 % CI: 0.77–0.88). In the validation cohort, the pooled c-index, sensitivity, and specificity for LR were 0.812 (95 % CI: 0.779–0.844), 0.83 (95 % CI: 0.76–0.88), and 0.68 (95 % CI: 0.63–0.73), respectively. In contrast, the pooled c-index, sensitivity, and specificity for ResNet were 0.821 (95 % CI: 0.788–0.853), 0.75 (95 % CI: 0.70–0.78), and 0.74 (95 % CI: 0.72–0.76) in the validation cohort, respectively. The analysis suggests that RF may demonstrate superior performance in the diagnosis of DKD. Further details are presented in Table 4.

4. Discussion

4.1. Summary of the main findings

Our *meta*-analysis shows that ML is relatively accurate in the prediction and diagnosis of DKD.

For forecasting the risk of DKD, the aggregated c-index was 0.832 (95 % CI: 0.803–0.862), with sensitivity of 0.79 (95 % CI: 0.74–0.84) and specificity of 0.79 (95 % CI: 0.74–0.83). In the diagnosis of DKD, the aggregated c-index, sensitivity, and specificity in the validation set were 0.82 (95 % CI: 0.73–0.89), 0.84 (95 % CI: 0.81–0.87), and 0.87 (95 % CI: 0.82–0.91), respectively. In the prediction of DKD, LR was predominantly used, showing comparable performance to other models. In the diagnosis of DKD, LR still prevailed, but algorithms such as ResNet, RF, and support vector machine (SVM) appeared to offer superior performance. This is the first systematic review and *meta*-analysis that concurrently summarizes the application of ML for both the prediction and diagnosis of DKD.

4.2. Prediction models for DKD

Proteinuria is a traditional biomarker for detecting the onset and progression of DKD[49]. However, since renal injury and repair are dynamic, proteinuria itself fluctuates dynamically. This non-linear progression results in uncertain trajectories of albuminuria. The sensitivity and specificity of proteinuria are limited in evaluating early-stage DKD[50]. The transition from diabetes to DKD is often insidious. By the time diabetes patients show urinary protein, changes in renal hemodynamics or even structural alterations may have already occurred. Interventions at this stage typically offer limited clinical benefits. Early detection of the risk of developing DKD in diabetes patients is important for the prevention and intervention of DKD, yet early identification of the risk of chronic kidney disease (CKD) in type 2 diabetes remains challenging[51]. A single biomarker like urinary protein or omics alone cannot accurately predict DKD. It has been found that integrating various risk factors to assess DKD risk is more reliable, advancing the prevention and management of DKD. Consequently, various risk prediction models for DKD have been developed[52–54]. A study by Roderick C. Slieker et al. identified 64 prediction models through systematic review and *meta*-analysis, and used external validation to verify the effectiveness of DKD prediction models. As artificial intelligence advances, ML has begun to be employed in model construction[55]. Wenhui Jiang et al.[56] performed a systematic review and *meta*-analysis, including 20 cohorts with 41,271 type 2 diabetes patients, to integrate risk factors and establish an ML model for early DKD prediction. This model, validated in an external cohort with a median follow-up of 2.9 years, demonstrated robust predictive performance for DKD, with an AUC of 0.765, sensitivity of 84.7 %, and specificity of 67.7 %.

To date, a systematic review and *meta*-analysis of ML applications in DKD has been published.[57], although the study has some limitations. First, it was geographically restricted to China, ultimately including 32 studies (both Chinese and English). Second, it did not present separate data for training and validation sets. Finally, it did not strictly differentiate between ML models for the prediction and diagnosis of DKD. In clinical practice, the prediction and diagnosis of a disease hold different significance. For the prediction of DKD, the focus is on identifying high-risk populations, so as to develop targeted prevention strategies. In the diagnosis of DKD, the aim is to minimize unnecessary biopsies. Our study strictly differentiates between ML tasks for the prediction and diagnosis of DKD. We included global studies on ML models for assessing the risk of DKD onset. These studies' data were from single-center, multicenter, and registry databases. The aggregated performance of ML prediction models was favorable, with an aggregated c-index of 0.810 (95 % CI: 0.790–0.831), sensitivity of 0.81 (95 % CI: 0.74–0.86), and specificity of 0.82 (95 % CI: 0.73–0.89), suggesting that ML is reliable for forecasting DKD.

In the prediction of DKD, LR was dominantly used, exhibiting good

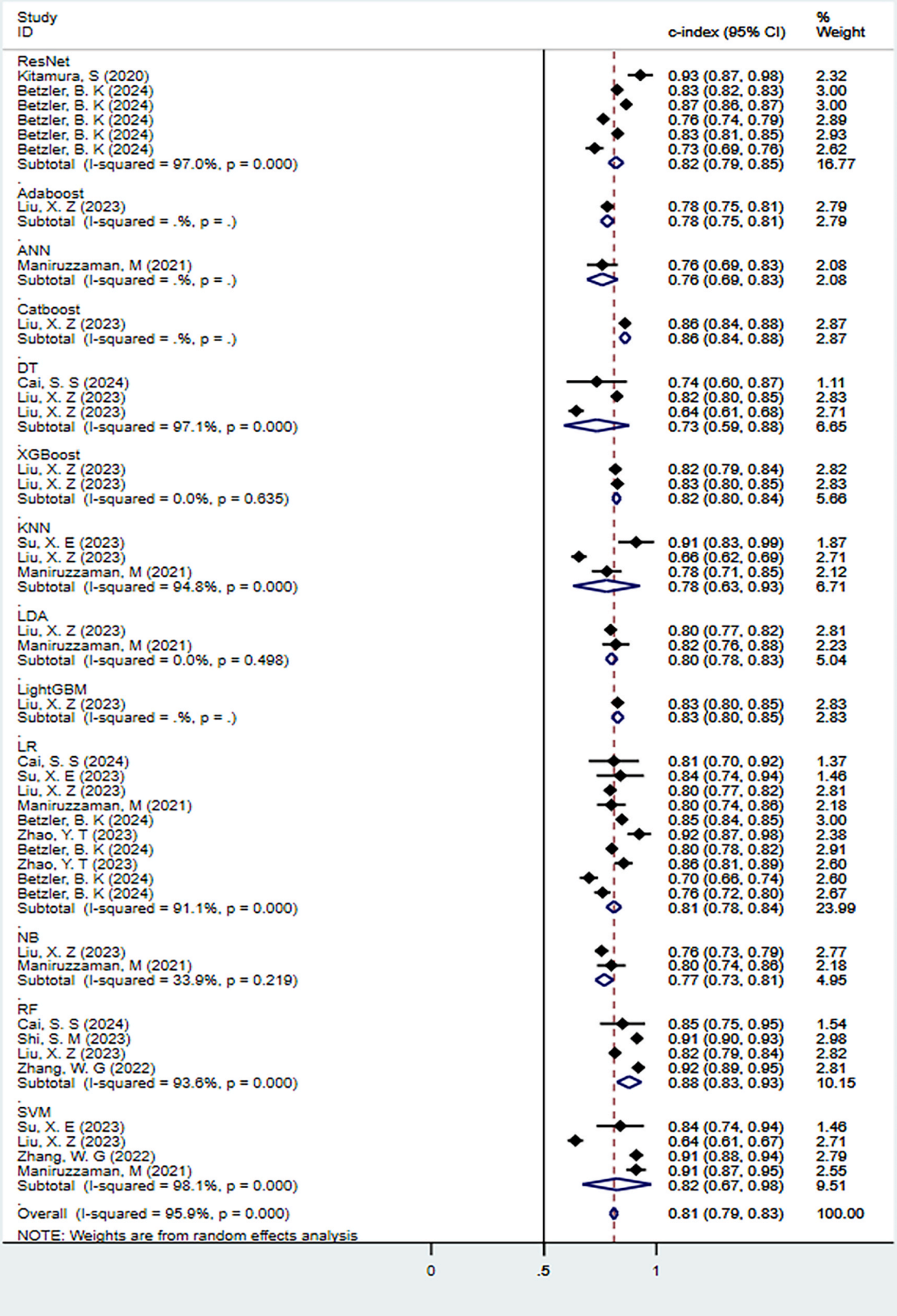


Fig. 5. Forest plot for c-index of models for the diagnosis of DKD in the validation set.

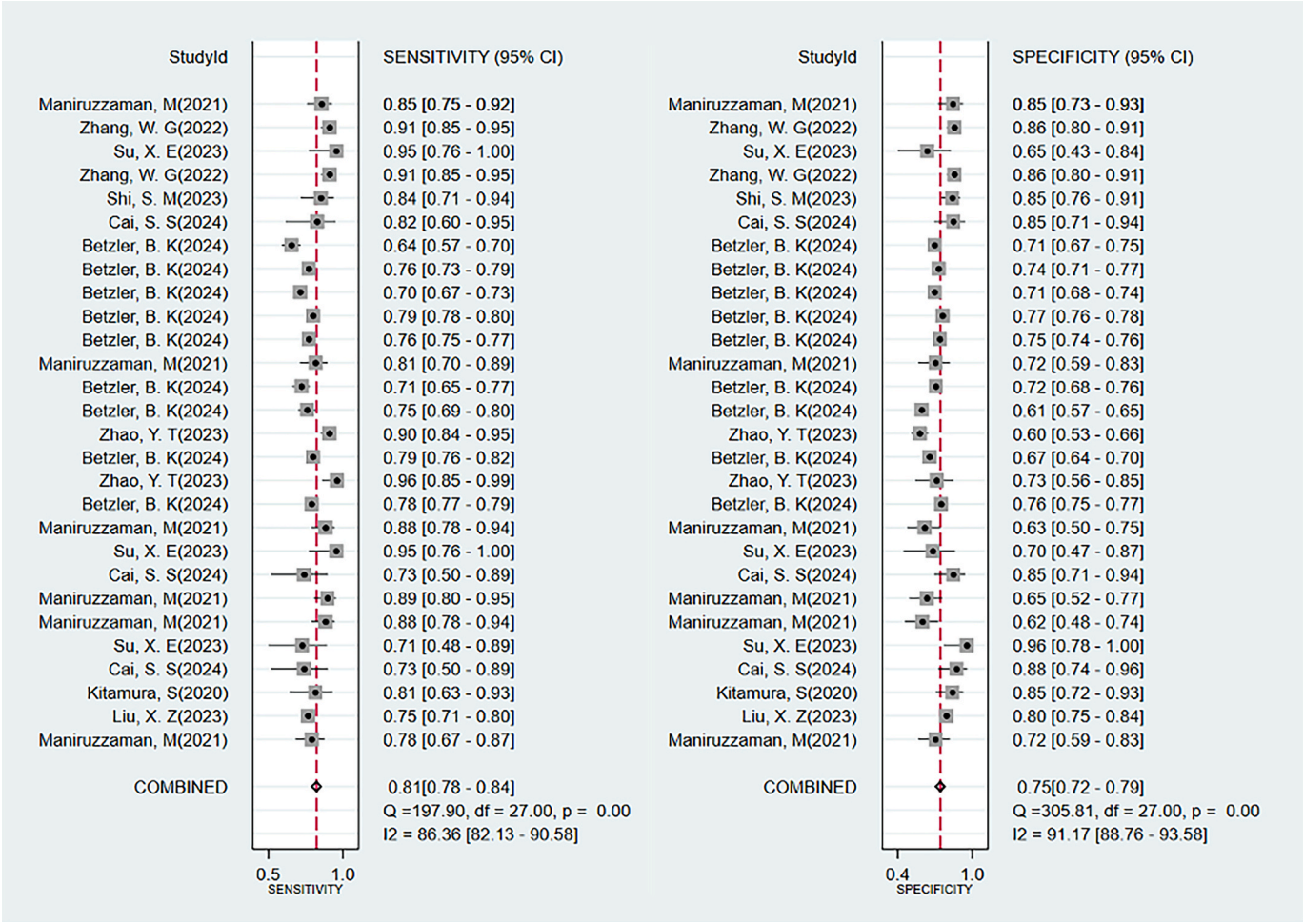


Fig. 6. Forest plot for the sensitivity and specificity of models for the diagnosis of DKD in the validation set.

Table 4
The pooled c-index of the models in the diagnosis of DKD.

No	Model	Training Set			Validation Set		
		Number	Sample size	c-index(95 %CI)	Number	Sample size	c-index(95 %CI)
1	ANN				1	133	0.760(0.692–0.828)
2	Adaboost				1	725	0.782(0.753–0.811)
3	Catboost				1	725	0.861(0.838–0.884)
4	DT				3	1513	0.735(0.591–0.879)
5	KNN	1	175	0.94(0.908–0.972)	3	902	0.779(0.631–0.926)
6	LightGBM				1	725	0.828(0.802–0.854)
7	XGBoost	1	359	0.965(0.950–0.980)	2	1450	0.823(0.804–0.841)
8	LDA				2	858	0.801(0.776–0.826)
9	LR	6	1008	0.904(0.873–0.935)	10	18,102	0.812(0.779–0.844)
10	NB				2	858	0.769(0.730–0.808)
11	RF	1	929	0.953(0.939–0.967)	4	1272	0.879(0.825–0.933)
12	SVM	2	1104	0.902(0.807–0.996)	4	1231	0.825(0.672–0.978)
13	ResNet				6	31,301	0.821(0.788–0.853)
14	Overall	11	3575	0.923(0.904–0.942)	40	59,795	0.812(0.794–0.829)

and stable performance. The pooled c-index of 10 LR models was 0.828 (95 % CI: 0.800–0.856). A *meta*-analysis of ML models for DKD prognosis [58] compared LR with non-LR models, but no evidence proved that non-LR models outperformed LR. However, this does not imply that LR is superior to all other ML models. One of the included studies[25] indicated that elastic net (EN) demonstrated strong performance for predicting the risk of DKD, with a c-index of 0.851 (95 % CI: 0.847–0.856). Another study[35] demonstrated that LightGBM also showed promising performance, achieving a c-index of 0.815 (95 % CI: 0.747–0.882). Nevertheless, due to limited studies on other ML models, their c-indices may not be representative.

4.3. *Diagnosis models for DKD*

In clinical practice, when diabetes patients are suspected of having DKD, clinicians often proceed with renal biopsy to confirm the diagnosis. However, as renal biopsy is an invasive procedure, it carries certain risks, and patients typically exhibit caution toward it, leading to poor compliance[6]. This has prompted clinicians to explore non-invasive diagnostic methods for DKD. An ideal diagnostic method

Table5
The pooled sensitivity and specificity of the models in the diagnosis of DKD.

No	Model	Training Set Number	sen(95 %CI)	spe(95 %CI)	Validation Set Number	sen(95 %CI)	spe(95 %CI)
1	ANN				1	0.78	0.715
2	Catboost				1	0.76	0.75
3	DT				1	0.71	0.88
4	KNN	1	0.89	0.81	2	0.7–0.88	0.62–0.95
5	XGBoost	1	0.85	0.97			
6	LDA				1	0.89	0.65
7	LR	5	0.84(0.75–0.90)	0.83(0.77–0.88)	9	0.83(0.76–0.88)	0.68(0.63–0.73)
8	NB				1	0.8	0.71
9	RF	1	0.84	0.9	3	0.81–0.91	0.71–0.85
10	SVM	2	0.84–0.85	0.80–0.91	3	0.85–0.96	0.67–0.86
11	ResNet				6	0.75(0.70–0.78)	0.74(0.72–0.76)
12	Overall	10	0.84(0.81–0.87)	0.87(0.82–0.91)	28	0.81(0.78–0.84)	0.75(0.72–0.79)

should be easy to measure, non-invasive, accurate, and repeatable. Metabolomics, a technique for studying multiple metabolites in body fluids, allows for the analysis of biochemical events within the organism and reveals complex metabolic networks. By identifying disease-specific metabolic profiles, metabolomics assists in disease diagnosis. As a result, metabolomics is regarded as a promising diagnostic tool for various diseases[59]. Research has applied metabolomics to DKD patients and found that significant changes in many metabolic pathways are highly correlated with the occurrence of DKD[60]. A review summarized several potential biomarkers associated with DKD, including amino acids, lipids, carbohydrates, the TCA cycle, and nucleic acids[61]. However, since metabolomics responds to genetic factors, lifestyle, and other variables, conflicting results have emerged across different cohorts. Moreover, high-throughput metabolomics is costly and time-consuming, making its routine clinical application currently unfeasible. Therefore, using metabolomics for DKD diagnosis remains challenging. With the advent of ML, several diagnostic models for DKD have been developed, though the evidence for their performance is scarce. To address this issue, this systematic review and *meta*-analysis was undertaken on ML-based diagnostic models for DKD. The present study included 40 models in the validation set with a total of 59,795 patients. A random-effects model was utilized to pool data. The pooled c-index, sensitivity, and specificity were 0.812 (95 % CI: 0.794–0.829), 0.81 (95 % CI: 0.78–0.84), and 0.75 (95 % CI: 0.72–0.79), respectively. The foregoing findings suggest that ML could be a potential auxiliary tool for diagnosing DKD.

Among the ML algorithms used in the diagnosis of DKD, LR remains the most commonly employed. Ten models were constructed utilizing LR, with a pooled c-index of 0.812 (95 % CI: 0.779–0.844). Six models were constructed utilizing the ResNet algorithm, achieving a pooled c-index of 0.821 (95 % CI: 0.788–0.853). Four models were constructed employing RF and SVM, respectively, with pooled c-indices of 0.879 (95 % CI: 0.825–0.933) and 0.825 (95 % CI: 0.672–0.978), respectively. ResNet, RF, and SVM algorithms demonstrated superior performance in the diagnosis of DKD.

LR is popular due to its simplicity and ease of interpretation[58]. However, LR struggles with large, multi-class feature sets and may suffer from underfitting when the feature space is large, leading to suboptimal performance. It is speculated that the inclusion of additional features in some diagnostic models may explain why LR does not outperform other algorithms. In contrast, ResNet, known for its strong learning capabilities and associative memory, is highly effective in approximating complex non-linear relationships and has been widely applied in fields such as computer vision[9]. Some models, such as ResNet, incorporate retinal images, which may explain why ResNet is second only to LR in model construction. However, ResNet typically requires a large number of parameters, contributing to the larger sample sizes observed in our ResNet validation sets. SVM remains a classic algorithm due to its high accuracy [62]. As a binary classifier, SVM can handle non-linear feature interactions, solve small-sample problems, and manage large feature spaces, making it well-suited for DKD diagnosis. However, SVM often demands significant memory and is less interpretable, presenting challenges in parameter tuning and computational efficiency. RF, consisting of an ensemble of random decision trees trained on bootstrapped subsets of the data, addresses these limitations. RF can deal with high-dimensional data, imbalanced datasets, and missing values, assess feature importance, and perform efficient training and prediction. Our study also showed that RF outperformed other algorithms[63]. Therefore, RF may have an advantage in the diagnosis of DKD.

4.4. Limitations of the study

Firstly, a large number of studies were included. However, the performance of the models varied, with some models reported in a small number of original studies, which could limit the interpretability of our findings. Secondly, the validation sets in the included studies were mainly generated from internally randomized samples, and independent external validation is lacking. Lastly, most of the included models did not account for the goodness-of-fit of the models.

5. Conclusions

Our study systematically summarizes various ML models applied to predict and diagnose DKD. The results indicate that ML shows promising performance in both predicting and diagnosing DKD. LR demonstrates stable performance in predicting the risk of DKD, while RF appears to be a better option for diagnosing DKD. Nevertheless, since ML is intricate, high heterogeneity exists among studies. Further guidelines are needed to enhance the performance of ML in clinical applications.

Author Contributions

All authors contributed to the study conception and design. Writing – original draft preparation: [Changmao Dai]; Writing – review and editing: [Changmao Dai and Xiaolan Sun]; Conceptualization: [Xueping Li]; Methodology: [Changmao Dai and Wei Chen]; Formal analysis and investigation: [Changmao Dai and Jia Xu]; Resources: [Changmao Dai and Maojun Chen]; Supervision: [Xueping Li],and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Ethics approval and consent to participate.

Not applicable.

Consent for publication

Not applicable.

Data statement.

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

Author Contributions.

All authors contributed to the study conception and design. Writing – original draft preparation: [Changmao Dai]; Writing – review and

editing: [Changmao Dai and Xiaolan Sun]; Conceptualization: [Xueping Li]; Methodology: [Changmao Dai and Wei Chen]; Formal analysis and investigation: [Changmao Dai and Jia Xu]; Resources: [Changmao Dai and Maojun Chen]; Supervision: [Xueping Li],and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Changmao Dai: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Xiaolan Sun:** Writing – review & editing, Formal analysis, Data curation. **Jia Xu:** Writing – review & editing, Software, Data curation. **Maojun Chen:** Writing – review & editing, Visualization. **Wei Chen:** Writing – review & editing. **Xueping Li:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmedinf.2025.105975.

Data availability

Data will be made available on request.

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